

Human Adipose Allograft Restores Volume Through Host-mediated Tissue Replacement

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Background: Advances in adipose tissue technology have led to the development of a commercial human adipose tissue allograft (hATA) for the restoration of adipose volume; hATA retains native adipose architecture, including structurally intact adipocytes, while eliminating immunogenic components to allow allogeneic use. This study evaluates the mechanisms by which hATA restores volume and the fate of donor-derived human deoxyribonucleic acid (DNA) and proteins following in vivo implantation.

Methods: Tissue structure and adipocyte integrity in hATA were evaluated by histology and free lipid analysis. The absence of extracellular damage-associated molecular patterns was assessed by DNA electrophoresis. In vivo remodeling and persistence of human-derived molecules were evaluated in a 12-week nude mouse model. Grafts were evaluated by visual appearance and weight; histology and immunohistochemistry; and the presence of human cells, adiponectin, and DNA. Systemic distribution was evaluated by human adiponectin detection in mouse serum.

Results: The hATA lacked immunogenic cells and extracellular damage-associated molecular patterns. Following in vivo implantation, hATA was populated by host-derived cells that remodeled and replaced the graft with host tissue. Human DNA was not detectable beyond 4 weeks, and human proteins progressively degraded. No human molecules were detected in systemic circulation.

Conclusions: Use of hATA provides immediate volume restoration through structurally intact adipocytes with retention maintained by host-mediated tissue regeneration. These findings support the graft replacement theory as the mechanism underlying hATA remodeling in vivo. (*Plast Reconstr Surg Glob Open* 2026;14:e7997; doi: 10.1097/GOX.0000000000007997; Published online 4 August 2026.)

INTRODUCTION

Restoration of lost adipose volume remains a significant clinical challenge, particularly when autologous fat grafting is not feasible or desirable. A human adipose tissue allograft (hATA) has recently been developed as an alternative to autologous fat grafting; hATA is composed of devitalized human adipose tissue particles suspended in a cryopreservation solution. It retains the native tissue architecture, including structurally intact adipocytes, while lacking immunogenic components, allowing its allogeneic use.¹ Available on demand with convenient long-term frozen or refrigerated storage, hATA is designed to address small

volume defects via minimally invasive implantation without the need for matching or immunosuppression.

Previously, clinical use of hATA in patients with pain and ulceration due to plantar fat pad loss has shown that restoration of the fat pad results in pain reduction and protection against reulcerations.²⁻⁴ The purpose of this study is to investigate the mechanisms underlying adipose volume restoration with hATA, as well as the in vivo fate of donor deoxyribonucleic acid (DNA) and other molecules following implantation.

MATERIALS AND METHODS

Human Adipose Tissue Procurement and Processing

Human adipose tissues were provided by the National Disease Research Interchange (Philadelphia, PA) and Zen-Bio, Inc. (Durham, NC) according to their approved

From the Britecye, Inc., Frederick, MD.

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protocols, with written informed consent obtained. Human adipose from cadaveric or lipoaspirate donors was processed and preserved as described in Regulski et al.¹ Two versions of human adipose allografts were generated for the study: devitalized hATA and viable hATA (vhATA). The hATA was produced using a small-scale laboratory process that mimicked the manufacturing process and composition of 2 commercial hATAs, Liposana and Lipoderma (Britecye, Inc., Frederick, MD). To devitalize the tissue before processing, human adipose was frozen in the absence of a cryopreservation solution. Stromal vascular fraction (SVF) cells and adipocytes were isolated from adipose by enzymatic digestion using collagenase type 2 (Worthington, Lakewood, NJ) as previously described.⁵ Cell and tissue viability was assessed by staining with ViaStain acridine orange/propidium iodide (Nextcelom Bioscience, Lawrence, MA) according to the manufacturer's protocol.

Lipopolysaccharide Assay

Tissue samples (0.5 mL tissue in 0.5 mL of Dulbecco Modified Eagle Medium [DMEM] supplemented with 10% fetal bovine serum [Life Technologies, Carlsbad, CA]) or cells isolated from 5 mL of tissue were incubated in 24-well plates in the presence of 10 µg/mL lipopolysaccharide (LPS) for tissues and 1 µg/mL LPS for cells for 48 hours. Wells with tissue and cells without LPS served as a control. After 48 hours, culture supernatants were collected and tested for the presence of tumor necrosis factor-α (TNF-α) using a human TNF-α kit for Luminex (Life Technologies) or an enzyme-linked immunosorbent assay (R&D Systems, Minneapolis, MN) according to the manufacturer's protocol.

Free Oil Measurement

The hATA samples were centrifuged in syringes at 400 relative centrifugal force for 10 minutes at room temperature. After centrifugation, tissue and oil layers were measured with a calibrated ruler in millimeters and expressed in percentages (%).

Histology and Immunohistochemistry

Tissue processing and staining were performed at Histoserv, Inc. (Germantown, MD). Samples were evaluated by a certified blinded pathologist. For immunohistochemical (IHC) staining, rabbit primary anti-mouse cluster of differentiation (CD) CD31 antibodies (1:100), antimouse Ki67 (1:100), and anti-human CD36 (1:400) from Cell Signaling Technology (Danvers, MA) were used.

Histology scores for graft remodeling were calculated as the sum of scores for vacuoles, cell infiltrate, and fibrosis; histology scores for new tissue formation were calculated as the sum of scores for new adipocytes, vascularization, and proliferating cells. The score scale ranged from 0 to 4 (not present, mild, moderate, and severe).

Cell Cultures and Adipogenic Differentiation

Adipogenic differentiation was performed as described in Choi et al.⁵ Cells isolated from hATA and vhATA at 4,

Takeaways

Question: How does human adipose tissue allograft (hATA), a commercially available alternative to autologous adipose and dermal fillers, restore lost tissue?

Findings: In vivo, hATA was populated by host-derived cells capable of adipogenic differentiation. Host cells remodeled and replaced hATA with newly formed host tissue over time. Donor-derived human cells, deoxyribonucleic acid, and adiponectin degraded at the implantation site and were not systemically distributed. Devitalized hATA retained volume comparable to viable grafts.

Meaning: Use of hATA provides immediate volume through structurally intact, lipid-filled adipocytes. Long-term effect is maintained through host-mediated tissue regeneration that replaces hATA, demonstrating that viable donor cells are not required for durable volume restoration.

8, and 12 weeks after implantation in mice were plated in a 24-well plate. When cells formed a monolayer, DMEM containing 10% fetal bovine serum was replaced with adipogenic differentiation medium (Life Technologies). Oil Red O (Sigma, St. Louis, MO) was used to detect differentiated cells.

Adipose Graft Extracts and Human Adiponectin Detection

Human adiponectin in mouse serum and in adipose graft extracts was measured by enzyme-linked immunosorbent assay (R&D Systems) using the manufacturer's protocol. Adipose extracts were prepared by mixing tissue and DMEM at a 1:1 (w/v) ratio in Eppendorf tubes. Tubes were vortexed and incubated on a rotator for 2 hours at room temperature, followed by centrifugation at room temperature for 10 minutes at 13,000 rpm. Extracts were transferred into new Eppendorf tubes and stored at -80°C until analysis.

DNA Analyses

DNA isolation was performed using a PureLink Genomic DNA kit (Invitrogen, Carlsbad, CA) according to the manufacturer's protocol. Extracted DNA was quantified using a NanoDrop ultraviolet spectrophotometer. DNA was analyzed using 1.2% Flash Gel DNA cassettes (Lonza Rockland, Rockland, ME) and a Flash Gel DNA marker (50–4000 bp) (Lonza Rockland) according to the manufacturer's protocol. Alu primers and Alu-qPCR (Applied Biosystems, Foster City, CA) were performed as described by Funakoshi et al.⁶

Nude Mouse Study

Animal studies were conducted at Noble Life Sciences, Inc. (Sykesville, MD) after protocol approval by the Noble Life Sciences Institutional Animal Care and Use Committee. Thirty 6-week-old female athymic mice (Envigo, Madison, WI) received a subcutaneous injection of either 0.5 mL hATA (n = 15) or vhATA (n = 15) via a 20G needle under general anesthesia. In this murine model, subcutaneously implanted grafts were anatomically

separated from native adipose tissue by the panniculus carnosus, allowing for clear identification of the implanted material during tissue harvest. Pain was controlled by medication. Animal health was monitored daily, and body weights were recorded weekly to assess volume retention. On the day of the injection, baseline blood serum samples were collected. At weeks 1, 2, and 4 postinjection, blood serum samples were collected for the detection of human adiponectin, and grafts were collected for DNA analysis. Gross evaluation and the weight of adipose grafts, as well as histology, IHC, and cellular analyses, were performed at weeks 4, 8, and 12.

Statistical Analysis

Statistical analyses were performed using GraphPad Prism and Microsoft Excel. Descriptive statistics (mean, SD, %, relative %) were calculated to summarize experimental data. Differences between groups were evaluated using an unpaired 2-tailed Student *t* test. *P* values of less than 0.05 were considered statistically significant.

RESULTS

hATA Lacks Immunogenic Cells and Extracellular Damage-associated Molecular Patterns

One of the major sources of immunogenicity in adipose tissue is blood cells present within the native vasculature. These cells were removed from the adipose tissue via multiple washing steps. (See figure, **Supplemental Digital Content 1**, which displays the photographs showing that tissue washes incorporated into adipose processing eliminate blood cells present in adipose. An example shows elimination of blood from adipose collected by lipoaspiration [top, incoming tissue; bottom, processed tissue], <https://links.lww.com/PRSGO/F129>.) Despite the removal of blood cells, the processed fresh tissue still contained resident immunogenic cells, as evidenced by TNF- α secretion in response to LPS stimulation. Isolation of adipocytes and SVF cells from the processed fresh tissue further demonstrated that resident immunogenic cells were localized within the SVF (Fig. 1).

Cryopreservation of the processed adipose significantly reduced, and devitalization completely eliminated, resident immunogenic cells (Fig. 1). Fluorescent staining confirmed that the processed cryopreserved viable adipose derived from both lipoaspirate and cadaveric donors contained viable SVF cells (Fig. 2). Cryopreserved adipose from lipoaspirate donors had approximately 50% viable adipocytes, whereas adipose from cadaveric donors had almost no viable adipocytes. As expected, devitalized cryopreserved adipose (hATA) had no viable cells (Fig. 2).

The hATA retained the native tissue architecture and adipocyte integrity required to keep intracellular lipids and damage-associated molecular patterns (DAMPs), such as DNA, within cells. Hematoxylin and eosin–stained histological images showed that there were no significant differences in adipose structure among unprocessed adipose, processed adipose, and cryopreserved hATA after thawing (Fig. 3). The minimal amount of free (extracellular) lipid

confirmed that although cells in hATA were devitalized, intracellular lipids remained inside the adipocytes. The amount of free lipids was less than 10% in all samples tested, including those exposed to 37°C, which simulates the postimplantation body temperature. (See figure, **Supplemental Digital Content 2**, which shows that free oil in hATA was measured after hATA centrifugation to separate the aqueous, tissue, and free oil layers [A]. B, The bar graph shows the percentage of free oil measured in hATA immediately after thawing and after 24 hours or 4 weeks of incubation at 37°C. The bar graph shows the mean $\pm$ SD for 10 experiments, <https://links.lww.com/PRSGO/F130>.) Similar to lipid retention, DNA remained intracellular (Fig. 4). The total amount of DNA measured in unprocessed adipose was 33–44 ng/mg of tissue and was 13 ng/mg of tissue in hATA. Agarose gel electrophoresis showed that high-molecular-weight DNA in hATA remained intracellular with only trace amounts of low-molecular-weight DNA, which would not be expected to trigger an inflammatory response if released extracellularly (Fig. 4).⁷ During exposure of hATA to 37°C, high-molecular-weight DNA was degraded intracellularly, with subsequent release of small DNA fragments (Fig. 4).

Host Cells Remodel and Replace hATA by a Newly Formed Tissue In Vivo

In vivo evaluation of hATA was performed using a nude mouse model. (See figure, **Supplemental Digital Content 3**, which displays a schematic representation of the in vivo testing of hATA and vhATA in a nude mouse model, <https://links.lww.com/PRSGO/F131>.) A version of adipose tissue allograft containing viable cells (vhATA) was also tested for comparison. Both hATA and vhATA were derived from a cadaveric donor and processed similarly, with the exception that hATA was devitalized, whereas vhATA was not. Implantation of human adipose grafts was well tolerated with no negative effects on animal health or behavior observed throughout the study duration (data not shown). Representative images of hATA and vhATA grafts at 4, 8, and 12 weeks postimplantation showed no significant differences in appearance between grafts. (See figure, **Supplemental Digital Content 4**, which displays the visual appearance of hATA and vhATA at 4, 8, and 12 weeks after in vivo implantation in nude mice. One representative image for each graft at each time point is shown, <https://links.lww.com/PRSGO/F132>.) Graft weights decreased by approximately 40%–45% at week 4 for both hATA and vhATA, with a more profound weight reduction observed for hATA. After 4 weeks, graft weights remained stable for the duration of the study, with no significant differences between hATA and vhATA (Fig. 5).

Histological analysis demonstrated remodeling of implanted grafts with simultaneous new tissue formation in both grafts. Large vacuoles, newly formed adipocytes, newly formed CD31-positive blood vessels, and Ki67-positive proliferating cells were present in both grafts simultaneously at 4 and 12 weeks postimplantation. No significant differences in remodeling or new tissue formation were observed between 4 and 12 weeks, indicating that both processes were still ongoing at week

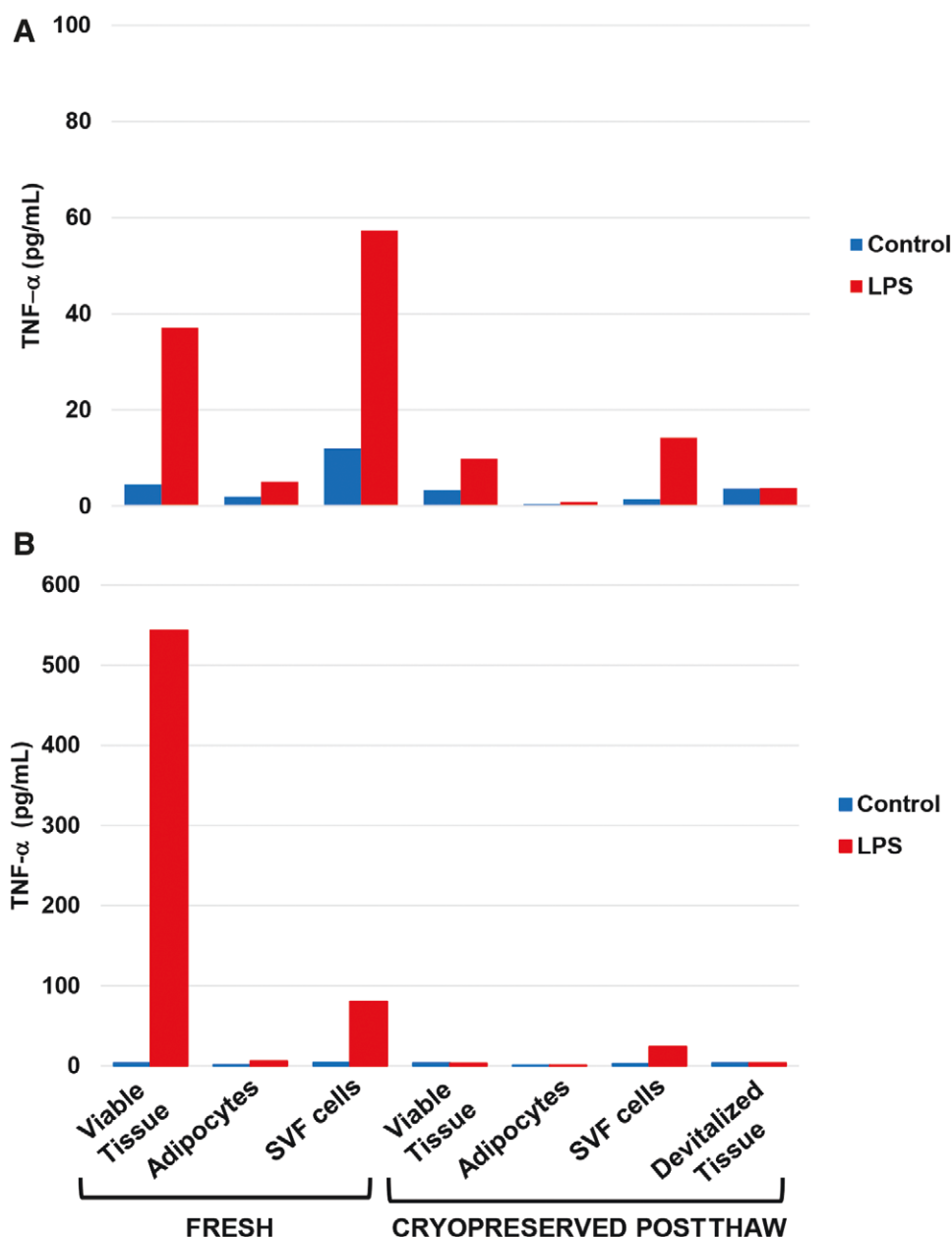


Fig. 1. Tissue processing and cryopreservation eliminate immunogenic cells in human adipose tissue. Bar graphs show results for lipoaspirate (A) and cadaveric (B) donors. The presence of immunogenic cells was detected using an LPS challenge assay, as described in Materials and Methods.

12. (See figure, **Supplemental Digital Content 5**, which displays hematoxylin and eosin–stained and IHC–stained mouse CD31 and Ki67 representative images of hATA and vhATA at 4 and 12 weeks after in vivo implantation in nude mice. Black circles indicate new adipocyte formation in the tissue; red arrows indicate mouse blood vessels in the human grafts; black arrows indicate Ki67–positive proliferating cells; asterisks indicate degrading adipocytes; scale bars: 100 μ M, <https://links.lww.com/PRSGO/F133>.)

Histological scores for graft remodeling and new tissue formation are shown in **Figures 6** and **7**, respectively. Remodeling of hATA started faster with a higher score at

week 4 compared with vhATA; however, no significant differences in remodeling scores were observed at later time points (**Fig. 6**). Additionally, no significant differences in new tissue formation were observed between hATA and vhATA at any time point (**Fig. 7**).

Further evaluation demonstrated progressive degradation of human-derived molecules within the implanted grafts. Human CD36, a receptor expressed on adipocytes, which was present before implantation, became undetectable 4 weeks postimplantation. (See figure, **Supplemental Digital Content 6**, which shows representative images of human CD36 IHC staining demonstrating that human CD36 was not detectable in hATA and vhATA grafts

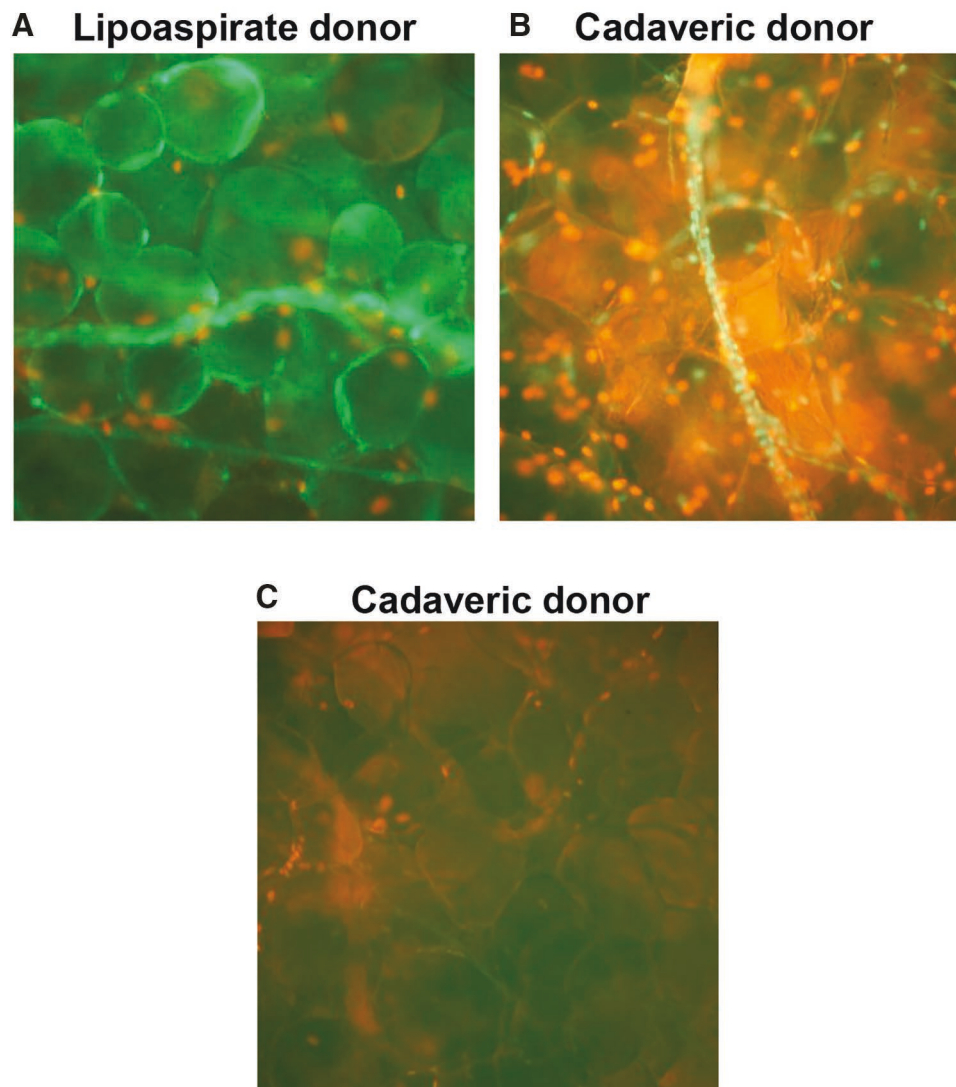


Fig. 2. Detection of viable cells in adipose tissue. Fluorescent images of acridine orange/propidium iodide-stained tissues confirm the presence or absence of viable cells in cryopreserved viable adipose tissue (A, B) and devitalized adipose tissue (C) after thawing.

4 weeks postimplantation in mice; scale bar: 100 μ M, <https://links.lww.com/PRSGO/F134>.)

Human DNA was not detectable beyond 4 weeks postimplantation (Fig. 8), and approximately 10% of human adiponectin remained within the grafts at 4 weeks (Fig. 9). Human adiponectin was not detected in mouse blood circulation at any time point (Fig. 9).

Both grafts were populated by host mouse cells, which were successfully isolated and cultured. The population of mouse cells isolated from implanted human grafts contained mesenchymal stem cells capable of adipogenic differentiation. (See figure, Supplemental Digital Content 7, which displays representative images of Oil Red O staining of hATA and vhATA isolated and culture-expanded control and adipodifferentiated cells, <https://links.lww.com/PRSGO/F135>.) The presence of viable human cells within the isolated cell population was detected by the secretion

of human adiponectin and expressed as the percentage of SVF cells in the tissue at day 0. As expected, no viable human cells were detected in hATA isolates. Only residual human cells (<0.0004%) were detected in vhATA isolates at 4 weeks postimplantation (Fig. 9), but not beyond 4 weeks (data not shown).

DISCUSSION

Recognizing the therapeutic potential of adipose tissue, a novel adipose processing technology has been developed that retains the native adipose structure while eliminating immunogenic components, thus enabling allogeneic use.¹ Adipose immunogenicity arises from 2 primary sources: resident immunogenic cells and intracellular DAMPs released from damaged cells during processing. Results of this study confirmed that hATA has neither immunogenic cells nor DAMPs (Figs. 1, 4). Devitalization

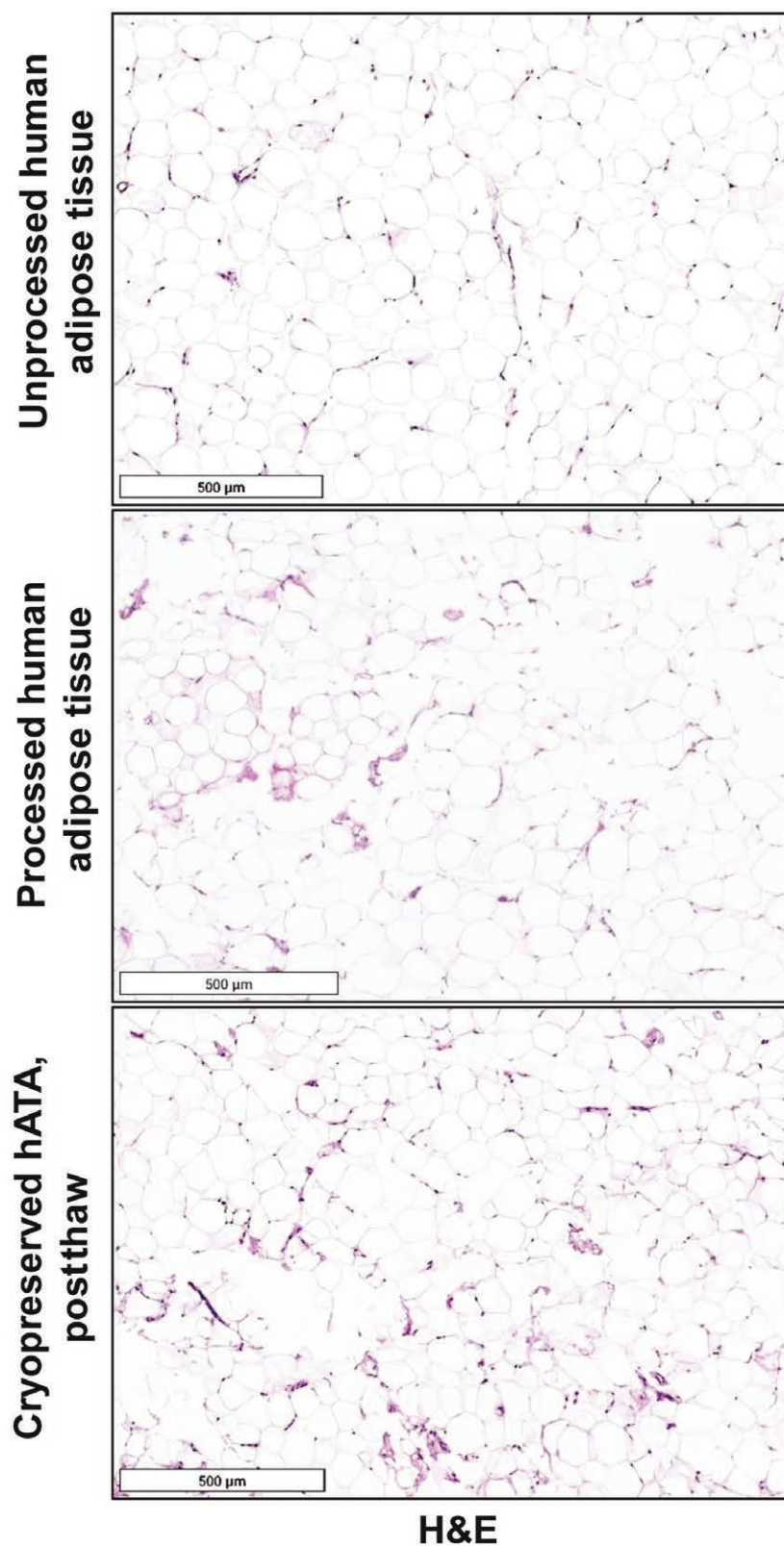


Fig. 3. Structure of adipose tissue. Images of hematoxylin and eosin (H&E)-stained adipose tissue sections show that hATA retains the architecture of native adipose and the structural integrity of adipocytes (scale bar: 500 μ m).

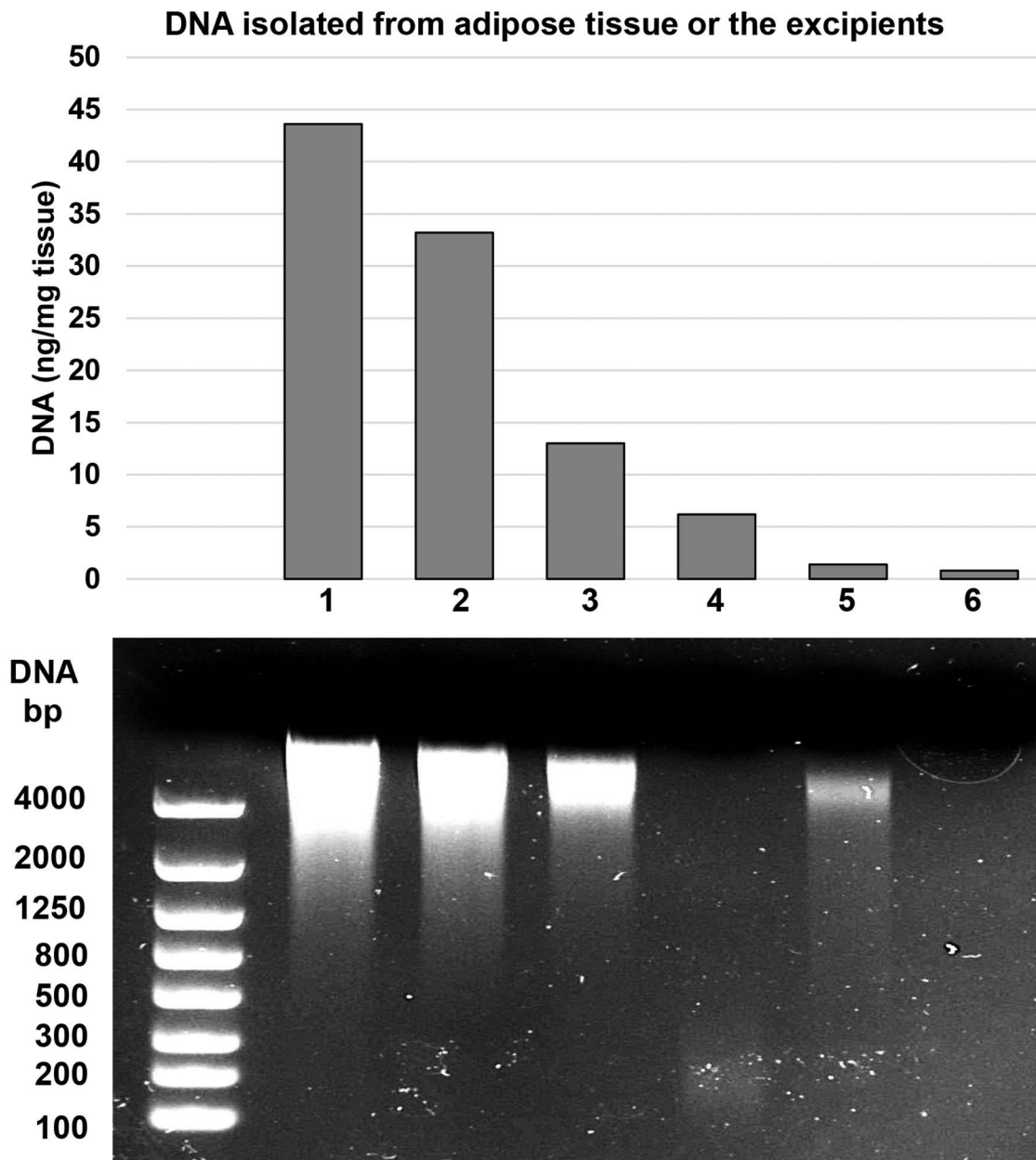


Fig. 4. DNA in adipose tissue samples. The bar graph shows the DNA content in unprocessed adipose tissue, hATA, and the hATA excipient. The agarose gel image shows DNA location in hATA. Results for 1 representative experiment are shown. Lanes: 1 and 2, unprocessed human adipose tissue; 3 and 4, DNA in hATA and its excipient after storage at -80°C ; 5 and 6, DNA in hATA and its excipient after incubation for 4 weeks at 37°C .

and decellularization are commonly used methods to overcome cellular immunogenicity.⁸ Although decellularization eliminates immunogenic cells, it disrupts cells and results in DAMP release, which increases immunogenicity.^{8–10}

High-molecular-weight DNA is a key DAMP that can activate innate immune responses, leading to inflammation.⁹ Therefore, a maximum DNA threshold of 50 ng/mg of tissue has been established for decellularized tissues to ensure that DNA is removed after cell destruction.¹¹ Cells in devitalized and viable tissue allografts are not destroyed, and DNA remains inside the cells. Evaluation of hATA demonstrated that the DNA content in hATA was 13 ng/

mg of tissue, which was lower than that in unprocessed adipose reported here (33–43 ng/mg of tissue; Fig. 4) and in the literature.¹² DNA reduction in hATA reflected effective removal of blood-derived cellular content during processing. Moreover, data confirmed that high-molecular-weight DNA in hATA remained intracellular (Fig. 4). DNA in hATA degrades inside the cells, with release of low-molecular-weight DNA fragments, which is the mechanism of intracellular molecule elimination from dead cells in the body.⁷

In vivo evaluation of hATA was performed in a nude mouse model, which is the most common model in

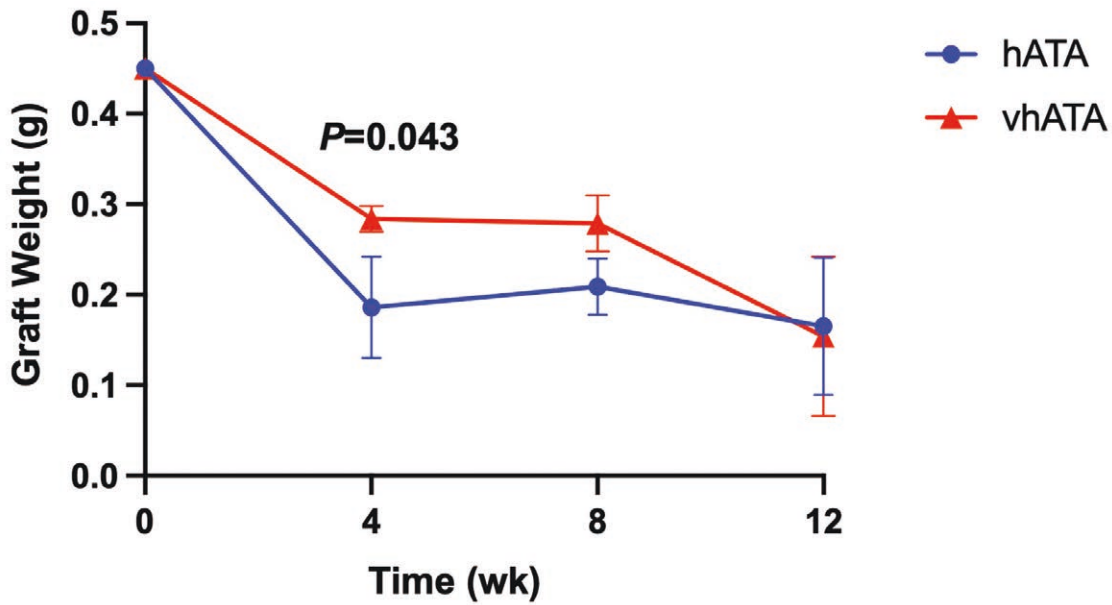


Fig. 5. Graft sizes of hATA and vhATA. The graph shows the weights of hATA and vhATA over time. Data are presented as mean $\pm$ SD graft weight in grams; N = 3 for each time point. A *P* value of less than 0.05 was considered statistically significant.

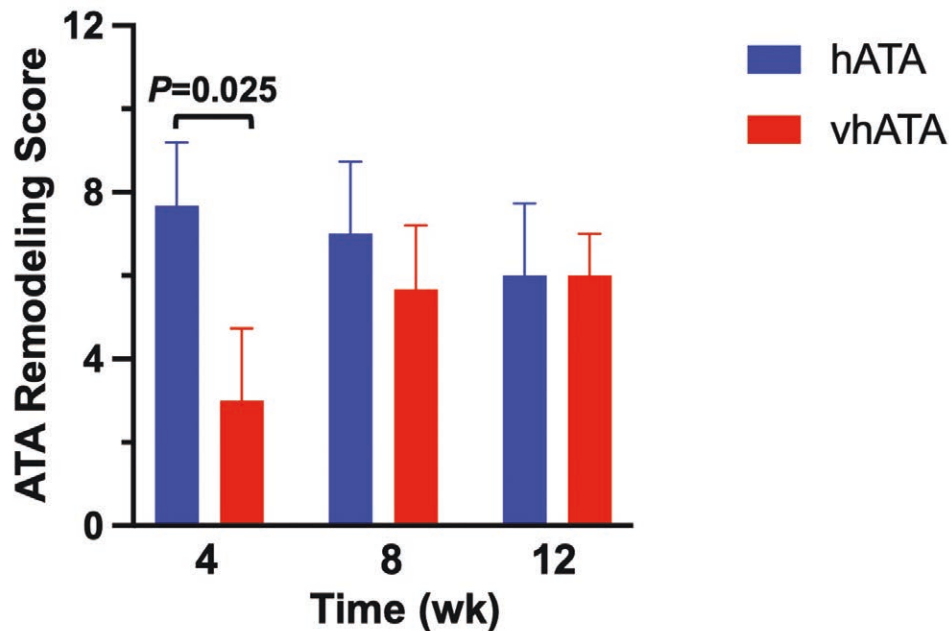


Fig. 6. Adipose graft remodeling. The bar graph shows histological remodeling scores at 4, 8, and 12 weeks. The graph shows mean $\pm$ SD scores. A *P* value of less than 0.05 was considered statistically significant.

adipose grafting research.¹³ A vhATA was included in the study as a control. After an expected initial weight reduction at week 4 due to absorption of the cryopreservation solution (~40%) in the hATA formulation, both grafts maintained stable volume retention of approximately 55%–60% at 8 and 12 weeks (Fig. 5). Histological analysis demonstrated concurrent remodeling and new tissue formation in both grafts, mediated by host cells and ongoing

at the final 12-week time point (Figs. 6, 7; Supplemental Digital Content 5, <https://links.lww.com/PRSGO/F133>).

Although several human adipose research studies exist, only 1 was identified that evaluated devitalized adipose. Kokai et al¹⁴ observed that graft retention in nude mice correlated with graft viability, whereas the present study shows similar graft retention between hATA and vhATA (Fig. 5). Differences in graft preparation, particle

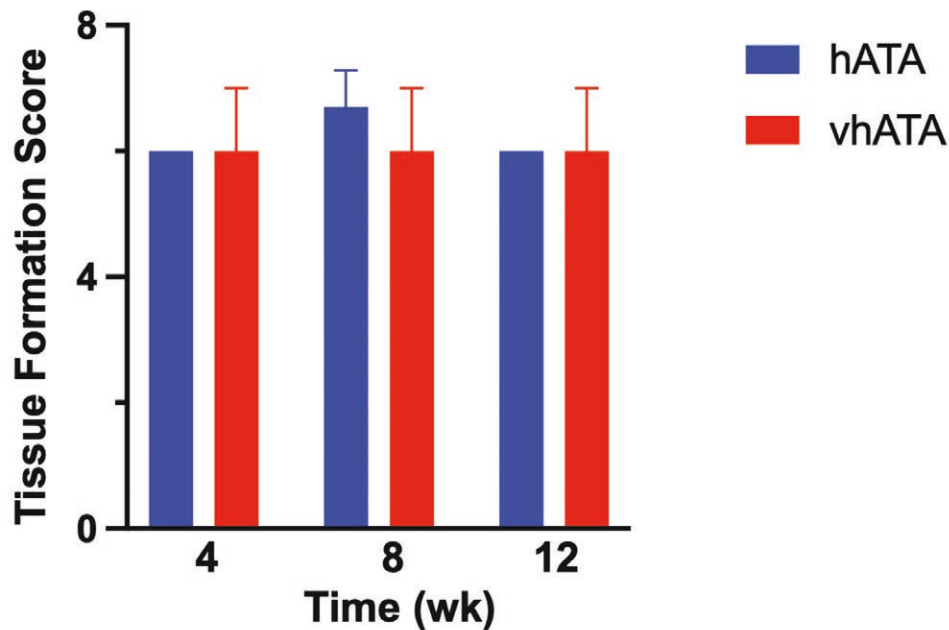


Fig. 7. New tissue formation in adipose grafts. The bar graph shows histological scores for new tissue formation at 4, 8, and 12 weeks. The graph shows mean $\pm$ SD scores. A *P* value of less than 0.05 was considered statistically significant.

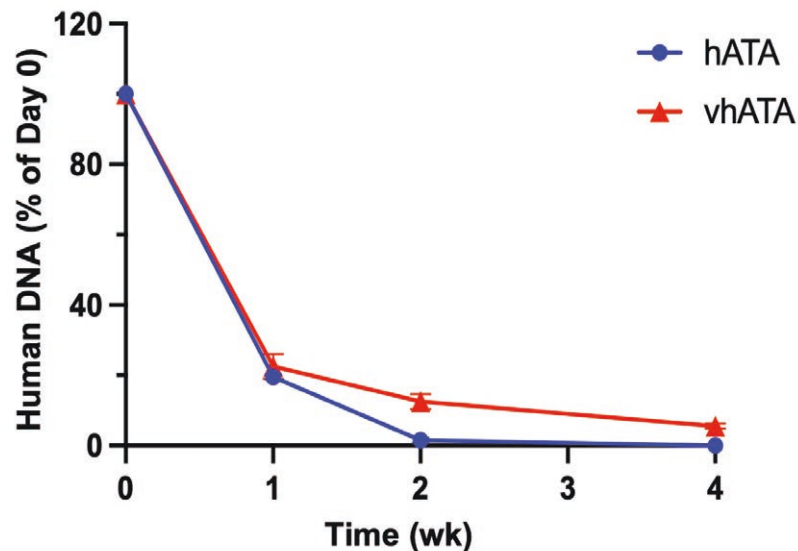


Fig. 8. Human DNA detection in grafts. The graph shows that DNA in hATA and vhATA rapidly degrades after in vivo implantation. The graph shows the mean $\pm$ SD of DNA expressed as a percentage of day 0.

size, implantation site, and technique may account for discrepancies between the 2 studies.^{15–18}

Fat graft retention in animals is typically assessed using ex vivo weight measurements, which require euthanasia of the study animal.¹³ Noninvasive imaging modalities, such as micro-computed tomography and magnetic resonance imaging, have been used to quantify graft retention over time without euthanizing animals¹³; however, their widespread use has been limited by the cost of equipment.

The use of digital and smartphone-based technologies has revolutionized noncontact measurements of skin changes in wound care, aesthetic, and reconstructive clinical practice.^{19–21} Although their application in preclinical animal models remains limited, continued advancements in these technologies may enable more accessible, objective, and standardized measurements in future animal studies.

The persistence and biodistribution of human adipose grafts in vivo were assessed via CD36, DNA, and adiponectin.

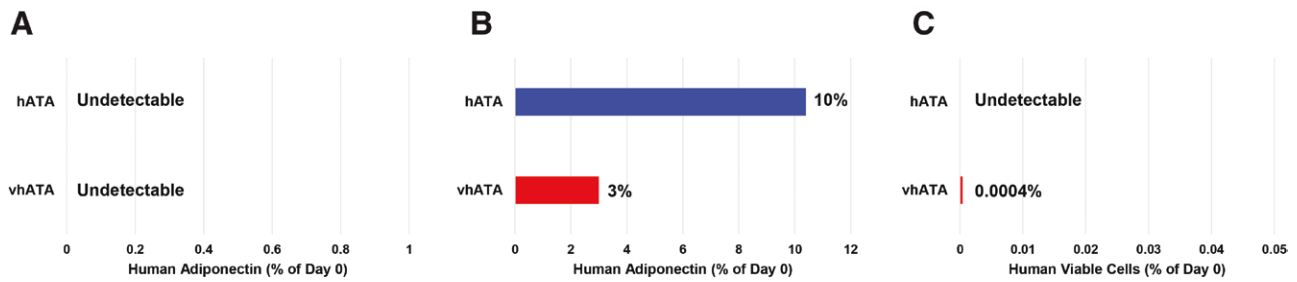


Fig. 9. Human adiponectin detection. Bar graphs show human adiponectin levels in mouse serum at 1 week after in vivo implantation of adipose grafts (A); human adiponectin levels in adipose grafts at 4 weeks after in vivo implantation (B); and the number of human culture–expanded, adipocyte-differentiated viable human cells isolated from adipose grafts at 4 weeks after in vivo implantation (in % relative to viable SVF cells in vhATA on day 0, before in vivo implantation) (C).

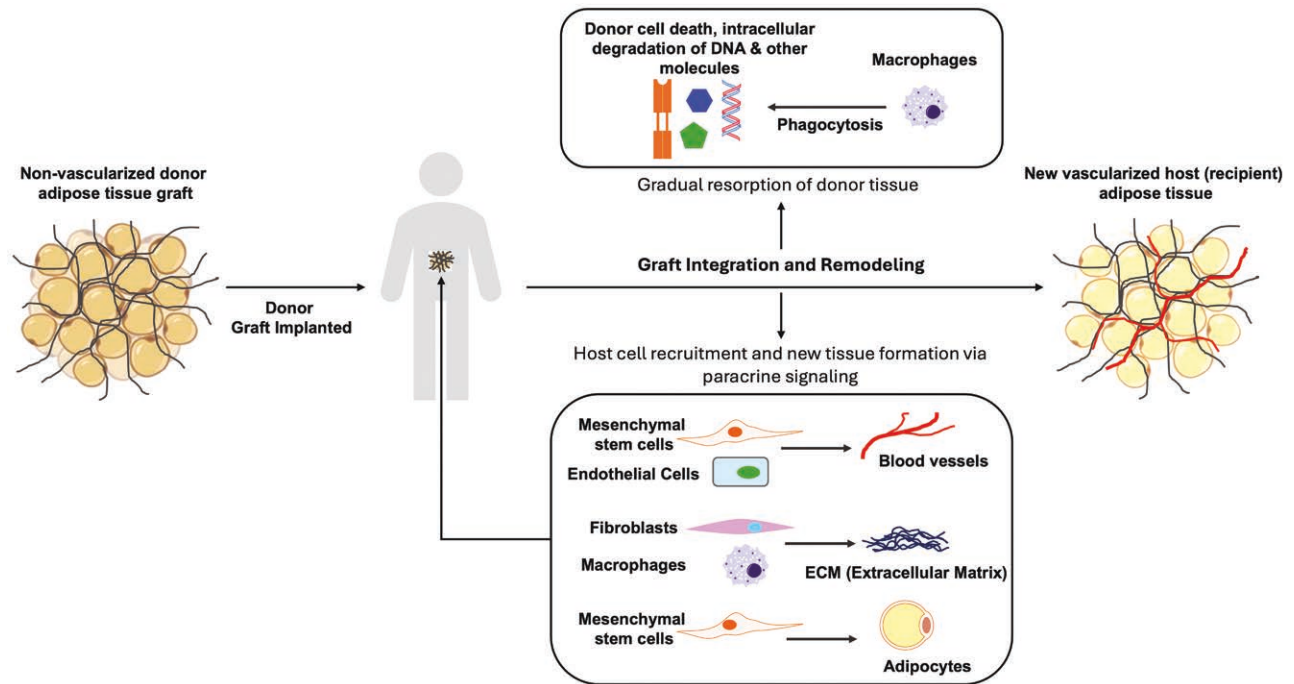


Fig. 10. The hATA graft replacement mechanism.

Human adipocytes are highly positive for the CD36 surface receptor expression, which became undetectable 4 weeks postimplantation (Supplemental Digital Content 6, <https://links.lww.com/PRSGO/F134>), similar to human DNA (Fig. 8). Both grafts were populated by host-derived cells, including mesenchymal cells that can differentiate into adipocytes (Supplemental Digital Content 7, <https://links.lww.com/PRSGO/F135>), indicating replacement of human adipose with mouse adipose. Using a highly sensitive adiponectin immunoassay, residual viable human cells were detected in vhATA isolates at 4 weeks, but not beyond (Fig. 9). Further biodistribution studies showed that less than 10% of human adiponectin remained within grafts at 4 weeks postimplantation (Fig. 9), and no circulating human adiponectin was detected in mouse serum, indicating no systemic effect (Fig. 9). Adiponectin is present at high levels in adipose tissue,^{22,23} making it an appropriate biodistribution marker.²⁴

These findings are consistent with published data for other nonvascularized tissue allografts. Devitalized allogeneic cartilage has been shown to exhibit significantly reduced immunogenicity while retaining osteoinductive properties comparable to those of viable autologous tissue.²⁵ Similarly, implantation of viable patellar tendon and anterior cruciate ligament allografts in animals resulted in rapid replacement of donor cells by host cells, with detectable allogeneic DNA persisting for up to 4 weeks; this replacement of the donor's cells was not due to host-mediated immune rejection.²⁶

Currently, there are 2 other commercial hATAs available in the United States: an adipose allograft matrix (Renuva, MTF Biologics, Edison, NJ) and a structural adipose filler (alloClac, Tiger Aesthetics, Franklin, WI).^{27,28} Implantation of these allografts in nude mice results in graft infiltration by host cells with neoadipogenesis and neovascularization within 3–4 weeks.^{27,28}

Retention of devitalized adipocytes in hATA confers 2 functional benefits. First, they provide volume and cushioning to the underlying tissues and organs immediately upon implantation.¹ Second, nonviable adipocytes serve as targets for macrophage infiltration, which is critical for revascularization and subsequent graft retention.^{29–31} Previously published hATA preclinical data in immunocompetent rats showed M2 macrophage infiltration and neovascularization within 3 days postimplantation in vivo.¹ This occurred significantly earlier than reported for other adipose allografts (3–4 weeks).^{27,28}

The graft replacement theory postulates that most adipocytes undergo ischemic death and will be replaced via differentiation of surviving stromal cells into adipocytes. The balance between regeneration and degeneration determines the final volume retention after fat grafting.^{32–34} This study supports and extends the graft replacement theory by demonstrating that even fully devitalized grafts undergo host-mediated remodeling and regeneration (Fig. 10). Thus, viable allogeneic donor cells may not be required for long-term volume restoration.

Devitalized human tissue allografts derived from cadaveric donors have a long history of clinical uses, with more than 1 million implantations performed annually in the United States alone.^{8,35} Because hATA became available only recently, additional time is required to accumulate clinical evidence to gain broad acceptance and use of hATA for restoration of lost tissue volume, which traditionally is achieved with autologous adipose grafting.

Future studies should focus on the development of hATA implantation protocols for achieving the best clinical outcomes. The primary limitation of this study is the use of immunodeficient nude mice, which lack key immune cells present in humans. The impact of altered cytokine profiles and immune responses in this model on the mechanisms of fat graft remodeling remains unknown. In addition, human fat grafting in mice represents a xenotransplantation model, which does not fully reflect the allogeneic use of hATA in patients. These limitations may affect the translatability of the findings to human outcomes and should be considered when interpreting the results.

CONCLUSIONS

As a commercially available tissue allograft, hATA is an alternative to autologous adipose grafting for the restoration of adipose volume. It retains the native structure of adipose, including adipocytes, while eliminating immunogenic cells. Upon implantation in vivo, hATA is actively remodeled by host cells and replaced by host-derived tissue without systemic distribution of human proteins. These findings demonstrate that viable allogeneic donor cells may not be required for durable volume restoration. The findings of this study support the graft replacement theory previously proposed for autologous adipose grafting.

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DISCLOSURES

Danilkovitch, Saunders, and Tom are employees of Britecyte, Inc. Danilkovitch and Tom are board members of Britecyte, Inc. The content of this article was expressly written by the listed authors. No ghostwriters were used to write this article.

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